Math 261-001
Quiz 9, November 6

Name	CSU ID #

Choose and answer **exactly two** of the four problems. If you attempt more than two, only the first two will be graded. Write your final answers in the boxes provided. Only answers in the boxes will be graded. You may use the surrounding space for scratch work. Each problem is worth the same number of points.
Throughout, $\vec{N}$ is the specified unit normal and $\vec{T}$ is the unit tangent in the direction of traversal.

1. In each part, use Stokes' theorem to replace the requested integral by the **other side of the theorem**. State the **type of integral** (line or surface) and the **vector field being integrated**. Do not compute derivatives or evaluate either integral.

   (a) Let $\vec{F} = (y, z, x)$, and let S be the outward-oriented upper hemisphere $x^2 + y^2 + z^2 = 4$, $z \geq 0$. The requested integral is
   $$\iint_S (\operatorname{curl} \vec{F}) \cdot \vec{N}\, d\sigma.$$

   (b) Let $\vec{G} = (z, x, y)$, and let C traverse the triangle's straight edges in the order
   $$(2, 0, 0) \longrightarrow (0, 2, 0) \longrightarrow (0, 0, 2) \longrightarrow (2, 0, 0).$$
   The requested integral is $\oint_C \vec{G} \cdot \vec{T}\, ds$.

(a) (b)

2. Let S be the annular surface
   $$1 \leq x^2 + y^2 \leq 4, \qquad z = 0,$$
   with upward normal. Give the equations of its boundary curves and the induced direction of each, viewed from above.

Outer boundary Inner boundary

3. Find the flux of $\vec{F} = (y, x, y + z)$ through the portion of $2x + y + z = 2$ in the first octant, oriented **upward**:

$$\iint_S \vec{F} \cdot \vec{N}\, d\sigma.$$

4. Let

$$\vec{F} = (z - x^{2026},\ x - y^{2026},\ y - z^{2026}),$$

and let C traverse the triangle's straight edges in the order

$$(2, 0, 0) \longrightarrow (0, 2, 0) \longrightarrow (0, 0, 2) \longrightarrow (2, 0, 0).$$

Use Stokes' theorem to evaluate $\oint_C \vec{F} \cdot \vec{T}\, ds$.